

HELLO EVERYONE

We will start @ 09:40pm

Number System



↳ Decimal NS:- $[0-9]$ $(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)$

↳ Base $\rightarrow 10$

$$\begin{aligned} \rightarrow 945 &\Rightarrow 9 \times 10^2 + 4 \times 10^1 + 5 \times 10^0 \\ &\Rightarrow 900 + 40 + 5 \Rightarrow \boxed{945} \end{aligned}$$

↳ $(945)_{10}$

[Bit Manipulation]

↳ Binary Number System

↳ [0-1] (0,1)

↳ Base = 2

Ex:- $\overset{5}{101}$, $\overset{7}{111}$, $\overset{8}{1000}$

↳ $(111)_2$

$$\begin{aligned} 101 &\Rightarrow 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ &\Rightarrow \underbrace{4}_{4} + 0 + 1 = (5) \end{aligned}$$

↳ Octal Number System

↳ Digits - $[0-7]$ (0, 1, 2, 3, 4, 5, 6, 7)

↳ Base - 8

↳ $(123)_8$

↑
Octal NS.



$$1 \times 8^2 + 2 \times 8^1 + 3 \times 8^0$$

↳ Hexadecimal N.s.

↳ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

↳ Base $\rightarrow 16$

↳ (0A2)₁₆

↳ Decimal to Binary → (Positive num.)

$$a = (56)_{10} \rightarrow (?)_2$$

⇒

$$(111000)_2$$

quotient = a

rem.

2	56	0
2	28	0
2	14	0
2	7	1
2	3	1
2	1	1
	0	

↑

ans is bottom to top.

$$\rightarrow (91)_{10} \rightarrow (?)_2$$

$$\rightarrow (123)_{10} \rightarrow (?)_2$$

$$\rightarrow (67)_{10} \rightarrow (?)_2$$

$$\rightarrow (110)_{10} \rightarrow (?)_2$$

2	123	1
2	61	1
2	30	0
2	15	1
2	7	1
2	3	1
2	1	1
2	0	

$$(1111011)_2$$

2	91	1
2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
2	1	1
2	0	

$$(1011011)_2$$

2	110	0
2	55	1
2	27	1
2	13	1
2	6	0
2	3	1
2	1	1
2	0	

$$(1101110)_2$$

$$\uparrow$$

2	67	1
2	33	1
2	16	0
2	8	0
2	4	0
2	2	0
2	1	1
2	0	

$$(1000011)_2$$

$$6 \ 5 \ 4 \ 3 \ 2 \ 1 \ 0$$

$$1 \times 2^6 + 0 + 1 \times 2^4 + 1 \times 2^3 + 0 + 1 \times 2^1 + 1 \times 2^0$$

$$64 + 0 + 16 + 8 + 0 + 2 + 1 = (91)_2$$

$$6 \ 5 \ 4 \ 3 \ 2 \ 1 \ 0$$

$$1 \times 2^6 + 1 \times 2^5 + 0 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0$$

$$64 + 32 + 0 + 4 + 2 + 0$$

$$(110)_{10}$$

$n = 56$ rem $\rightarrow$ ans.

$n \rightarrow$	2	56	0
$n \rightarrow$	2	28	0
$n \rightarrow$	2	14	0
$n \rightarrow$	2	7	1
$n \rightarrow$	2	3	1
$n \rightarrow$		1	1
$n \rightarrow$		0	

$$n = n/2; \Rightarrow n/=2;$$

String $ans = ""$
 $\hookrightarrow ans = "000111"$

reverse (ans) $\rightarrow$ 111000

~~56~~ ~~28~~ ~~14~~ ~~7~~ ~~3~~ ~~1~~ ~~0~~

```

while (n > 0) {
    int rem = n % 2;
    ans += (rem + "");
    n /= 2;
}

```

$\rightarrow$
 $rem + ans;$
 $\downarrow$
 ans
 $\rightarrow$ 111000

Decimal to binary (2)

Decimal to any base (b)
↑

$$(1234)_{10} \rightarrow (?)_8$$

b	56	—
	—	

2	56	0
2	28	0
2	14	0
2	7	1
2	3	1
2	1	1
	0	

8	1234	2
8	154	2
8	19	3
8	2	2
	0	

$$(2322)_8$$

$\hookrightarrow \%11 \rightarrow [0 - \overset{\nearrow 1,0}{10}]$

$\rightarrow [0-9]$

$\rightarrow [A]$

$\hookrightarrow \%12 \rightarrow [0-9]$
 $\rightarrow [A, B]$

$\hookrightarrow \%13 \rightarrow [0-9]$
 $[A, B, C]$

$\hookrightarrow \%14 \rightarrow [0-9]$
 $\rightarrow [A, B, C, D]$

$b \rightarrow 1-10 \rightarrow [0-9]$

$b > 10 \rightarrow [A$

$b \rightarrow (10)$

$\overset{\uparrow\uparrow\uparrow}{4710408}$

$\overset{\uparrow}{47A408}$
 $\uparrow\uparrow\uparrow$

$\hookrightarrow \%15 \rightarrow [0-9]$
 $[A, B, C, D, E]$

$\hookrightarrow \%16 \rightarrow [0-9]$
 $[A, B, C, D, E, F]$

$rem = 10 \rightarrow \textcircled{A}$ $12 \rightarrow \textcircled{C}$
 $11 \rightarrow B$ $13 \rightarrow D$

rem

ans = (rem + "") + ans

$10 \rightarrow A$
 $11 \rightarrow B$
 $12 \rightarrow C$
 $13 \rightarrow D$
 $14 \rightarrow E$
 $15 \rightarrow F$

if (rem == 10) $\rightarrow$ return A
if (rem == 11) $\rightarrow B$
C
D
E
F

$$\text{rem} = 10$$

$$\begin{array}{c} \swarrow \quad \searrow \\ 'D' - 'A' = \text{integer} \\ \swarrow \quad \searrow \\ \text{ascii value subtraction} \end{array}$$

$$\begin{array}{l} 10 - 10 = 0 \rightarrow A \\ 11 - 10 = 1 \rightarrow B \\ 12 - 10 = 2 \rightarrow C \\ 13 - 10 = 3 \rightarrow D \\ 14 - 10 = 4 \rightarrow E \\ 15 - 10 = 5 \rightarrow F \end{array}$$

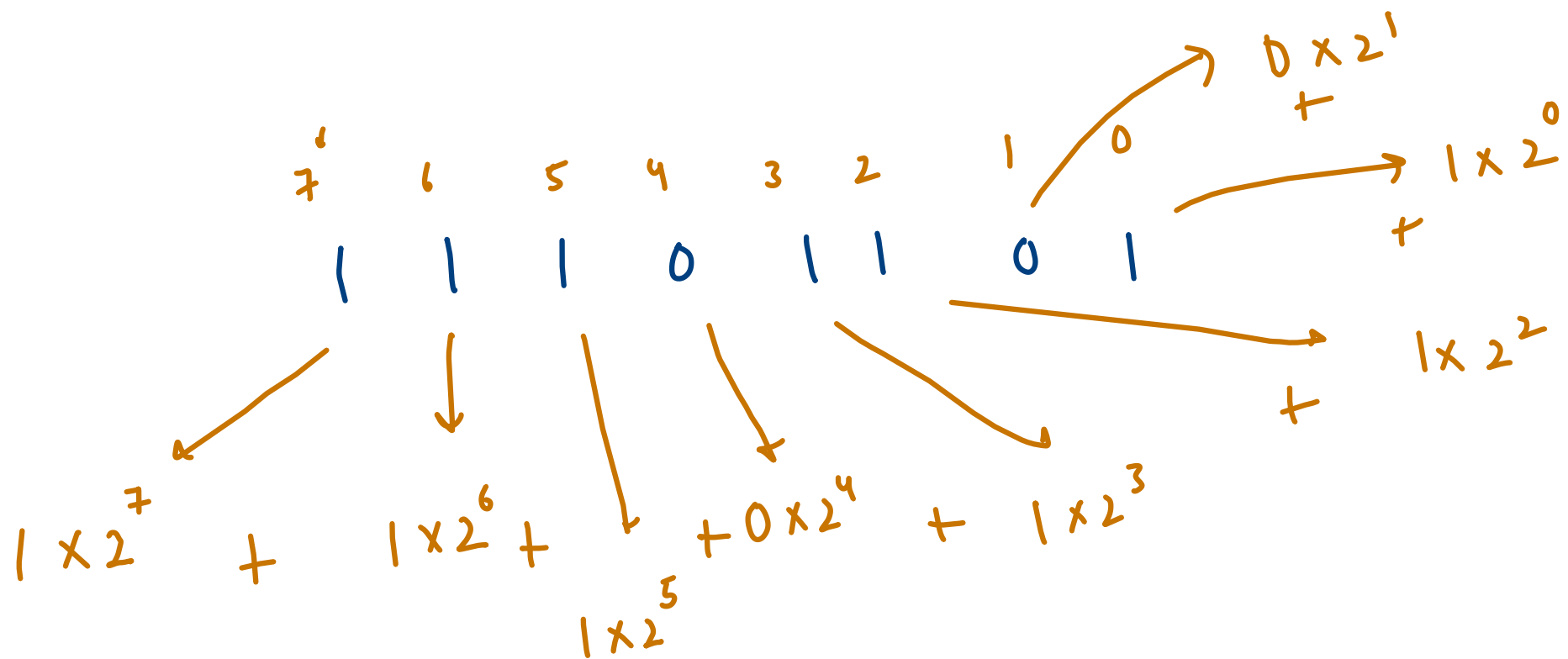
run 12

$$\begin{array}{c} = (3) \\ \downarrow \\ 2 + 'A' = \text{integer} \\ \uparrow \\ \text{ch} = (\text{rem} - 10 + 'A') \\ \downarrow \\ (69) \rightarrow \text{char} \quad (C) \end{array}$$

$$\begin{array}{c} \rightarrow (\text{rem} - 10 + 'A') \\ \downarrow \\ (\text{char}) (\text{integer}) \rightarrow (C) \end{array}$$

Binary to Decimal

$$\hookrightarrow (11101101)_2 \rightarrow (?)_{10}$$



$$128 + 64 + 32 + 0 + 8 + 4 + 0 + 1$$

$\rightarrow$ 237

$$\begin{array}{ccc} 0 & 1 & 2 \\ (1 & 0 & 1) \\ 2 & 1 & 0 \end{array} \xrightarrow{2} (?)_{10} \quad 2^0 = 1$$

$$\downarrow \text{pow} = 1 = 2^0 \times 2 = 2^1 \times 2 = 2^2 \quad '1' - '0' = (1 \times 1) + 0 \Rightarrow \text{ans} = 1$$

$$\text{ans} = 0$$

$$\begin{array}{ccc} 0 & 1 & 2 \\ (1 & 0 & 1) \\ 2 & 1 & 0 \end{array} \xrightarrow{2} \begin{array}{l} 1 \times 2^2 \\ \times 2^1 \\ 0 \times 2^0 \end{array} \quad \begin{array}{l} 1 \times 2^2 \\ \times 2^1 \\ 0 \times 2^0 \end{array}$$

$$4 + 0 + 1 = 5$$

$$\text{pow} = 1 \approx 2^0 \times 2 = 2^1 \times 2 = 2^2 \times 2 = 2^3$$

$$\text{ans} = 0 \quad \text{ans} + (1 \times 1) \Rightarrow 1 + 4 = 5$$

$$\text{ch} = '1' - '0' = (1 \times \text{pow})$$

$$\text{ch} = '0' - '0' = (0 \times 2^1)$$

$$\text{ch} = '1' - '0' = (1 \times 2^2) = 4$$